Replication Package for *Gendered Effects of the Minimum Wage*

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1 Overview

This replication package contains the data work and quantitative analysis for *Gendered Effects of the Minimum Wage*. It is organised around two empirical exercises and a quantitative model. The empirical analysis is written in Stata and uses the German Socio-Economic Panel (SOEP) and German regional data. The model is written in Fortran. We also provide the saved model results and MATLAB programs needed to recreate the model-based tables and figures without estimating and solving the model again.

The main folders have the following roles:

`SOEP/` Stata programs for the analysis based on individual and household data from the SOEP.

`Regional/` Stata programs for the regional analysis of the introduction of the German minimum wage.

`matlab/` Programs that read the supplied model results and prepare the model-based tables and figures.

`fortran/` Source code for estimating and solving the benchmark model and the three alternative models.

`benchmark/`, `fixwage/`, `FTonly/`, and `noDemoHet/` Saved model results used by the MATLAB programs.

`paper_outputs/` The tables and figures produced by the package, together with the tables that are supplied directly.

The code was prepared for Stata 18, MATLAB R2026a, and Intel oneAPI Fortran Compiler (`ifx`) 2026.0. The empirical exercises require data that we cannot redistribute. By contrast, the saved model results needed for the MATLAB analysis are included in the package.

2 Empirical analysis based on the SOEP

The SOEP analysis produces Tables 1 and 2 and Figures 12, 13, and 16. It also constructs the empirical moments used to estimate the benchmark model. Because the underlying microdata are subject to a licence, they are not included in the replication package. Users must obtain access to SOEP-Core v36 and the corresponding SOEP-CNEF files.

Place the following files directly in `SOEP/raw/`, keeping the original file names:

- `ppathl.dta`
- `pgen.dta`

- hbrutto.dta
- biobirth.dta
- artkalen.dta
- biocouplm.dta
- bdpequiv.dta
- bepequiv.dta

The folder SOEP/other/ contains the CPI series and the minimum-wage and minijob policy schedule used in the analysis. Once the restricted SOEP files have been added, open Stata from the root of the replication package and run:

```
do "SOEP/run_all.do" "C:/absolute/path/to/Replication_minwage_v2"
```

Replace the example path with the location of the package on your computer. The program prepares the working datasets and writes the final tables and figures to paper_outputs/. Intermediate data and the run log remain within the SOEP folder, with the log stored in SOEP/logs/.

3 Empirical analysis based on regional data

The regional analysis studies how employment changed across local labour markets with different exposure to the minimum wage. It produces Table 8 and the six panels of Figure 1, covering marginal, part-time, and full-time employment separately for men and women.

Create Regional/raw/ and place the following three files in that folder:

- county_kaitz_crosswalk.dta, a county-level file prepared by the user;
- arbeitsmarktregionen-2021.csv, which maps counties into the 2021 labour-market regions; and
- inkar_2021.csv, the INKAR regional-indicator extract.

The Kaitz index measures the bite of the minimum wage. It is the ratio of the statutory minimum wage of 8.50 euros to the median hourly wage in a region before the reform. The index used in the paper is constructed from the 2014 Structure of Earnings Survey and aggregated to German labour-market regions (*Arbeitsmarktregionen*). These source data cannot be distributed.

Users must prepare county_kaitz_crosswalk.dta from data to which they have authorised access. It should contain one numeric observation per county: the county identifier (kreise), the 2014 Kaitz index for the county's pre-reform labour-market region (kaitz_amr), and county population (pop_kreise_2015). Population is used to map the index into the 2021 labour-market regions.

After adding these files, open Stata from the root of the replication package and run:

```
do "Regional/run_all.do" "C:/absolute/path/to/Replication_minwage_v2"
```

The program builds the regional dataset, estimates the event-study regressions, and writes Table 8 and Figure 1 to paper_outputs/. Intermediate files and the run log remain within the Regional folder.

4 Using the saved model results in MATLAB

The supplied model outputs allow users to recreate the quantitative results in MATLAB without estimating or solving the Fortran model. The resulting tables and figures are written to `paper_outputs/`.

Start MATLAB with `matlab/` as the current folder and run:

```
run_all
```

The program reads the supplied results for the benchmark economy, the fixed-wage economy, the full-time-only economy, and the economy without demographic heterogeneity. It saves figures in EPS and PNG formats and tables in TeX format. The MATLAB program does not replace the empirical outputs made by Stata or the tables that are supplied directly in `paper_outputs/`.

5 Estimating and solving the model in Fortran

The Fortran code is intended for users who want to estimate and solve the model rather than rely on the saved results. There are four versions of the model:

- `fortran/benchmark/` contains the main model;
- `fortran/fixwage/` fixes the wage-posting function;
- `fortran/FTonly/` allows only full-time jobs; and
- `fortran/NoDemoHet/` removes demographic-type heterogeneity.

The Windows makefiles use Intel oneAPI `ifx` and Microsoft `nmake`. Initialise the Intel oneAPI environment, move to the folder of the model that you want to run, and use:

```
nmake /f makefile_win compile
nmake /f makefile_win run
```

The model reads inputs and writes results using paths relative to the selected model folder, so the executable should always be run from that folder. The settings that choose between estimation, solution, and counterfactual exercises are near the beginning of `src/main.f90`. After changing these settings, compile the program again. Full estimation and counterfactual exercises can take considerable time.

Each model writes its current results to its own `output/` folder. Users who want to prepare MATLAB figures from newly calculated results should first keep a copy of the supplied results and then copy their completed output folders to the corresponding top-level economy folder.

6 Tables and figures supplied with the package

The folder `paper_outputs/` is the common destination for the package. It contains Tables 1–11 and Figures 1–22, including all figure panels. Tables are stored as TeX fragments, while figures are supplied in EPS and PNG formats.

The two Stata exercises produce the empirical outputs: the SOEP programs make Tables 1 and 2 and Figures 12, 13, and 16, while the Regional programs make Table 8 and Figure 1. The MATLAB programs use the saved model results to make Tables 6 and 7 and the model-based figures. Tables 3–5 and 9–11 are supplied as completed tables because they combine estimated parameters, empirical moments, and selected model results.